

Dynamic Guideline: WHO-Based Maternal Health – Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

- **Guideline Title:** Meaningful youth engagement in a WHO guideline development process: case study of WHO consolidated guidelines for the prevention, diagnosis and treatment of postpartum haemorrhage
- **Publication Date:** 29 April 2026
- **Target Population:** Adolescents and young women (15–24 years) living in low- and middle-income countries; maternal health guideline developers; ministries of health; WHO staff and collaborating centres
- **Scope and Objectives:** This guideline (structured as a process evaluation and set of recommendations) provides an evidence-based framework for integrating meaningful youth engagement throughout the WHO guideline development lifecycle. It uses the development of the 2023 consolidated PPH guidelines as an instrumental case study. The primary objective is to enhance the responsiveness, equity, and acceptability of maternal health guidelines for young pregnant and postpartum individuals, thereby improving health outcomes and rights-based care.

1.2 Key Recommendations

Recommendation	GRADE Evidence Quality	Strength
1. A dedicated, diverse youth advisory panel should be convened at the start of every WHO maternal health guideline process and be embedded across all decision-making stages (scoping, evidence review, recommendation formulation, dissemination).	Very Low	Conditional

Recommendation	GRADE Evidence Quality	Strength
2. Youth-friendly participatory methods (digital storytelling, peer-led focus groups, anonymous online surveys) must be used to systematically collect values, preferences, and lived experiences of adolescent and young mothers, especially from underserved populations.	Very Low	Conditional
3. At least one youth representative with full voting rights should sit on the Guideline Development Group (GDG) and related sub-committees, and their participation must be supported by mentorship and accessible meeting materials.	Low	Strong
4. A youth-engagement monitoring and evaluation framework (including quantitative and qualitative indicators) should be built into the guideline dissemination and implementation plan, with results published in a publicly available case-study repository.	Very Low	Conditional

1.3 Guideline Development Methodology

- **Evidence Review Process:** A scoping review of frameworks for public and patient involvement in guideline development (2009-2025) was combined with a systematic review of qualitative studies and programme evaluations on youth engagement in sexual and reproductive health policy. Case-study data were collected through document analysis and in-depth interviews with GDG members, youth panelists, and WHO secretariat.
- **Consensus Method:** A modified Delphi process (two rounds) with an international panel of 35 experts (youth advocates, methodologists, clinicians, and maternal-health programme managers) was followed by a two-day virtual consensus conference. Recommendations were finalised by anonymous voting.
- **Conflict of Interest Management:** All participants submitted Declaration of Interest forms. One panel member with a financial interest in a youth-led maternal health NGO was restricted from voting on recommendations related to funding mechanisms; no other conflicts precluded participation.

II. Evidence Summary After WHO Latest Guideline Release (Consolidated guidelines for the prevention, diagnosis and treatment of postpartum haemorrhage) (Evidence Cutoff Date: 29 April 2026)

Note: The “WHO Latest Guideline Release” refers to the clinical guideline **WHO consolidated guidelines for the prevention, diagnosis and treatment of postpartum haemorrhage (published 15 March 2023)**. The youth-engagement case study summarised in Section I is a process document released on 29 April 2026 and does not alter the clinical guidance. Section II therefore synthesises all relevant new evidence that has emerged since the 2023 PPH guideline, up to 29 April 2026.

2.1 New Evidence Overview

- **Search Strategy:** Systematic literature search for original research, systematic reviews, and meta-analyses related to prevention, diagnosis, or treatment of postpartum haemorrhage.
- **Databases Searched:** MEDLINE (PubMed), Embase, Cochrane Central Register of Controlled Trials (CENTRAL), WHO Global Index Medicus, ClinicalTrials.gov, and the WHO International Clinical Trials Registry Platform (ICTRP).
- **Inclusion Criteria:** Randomised controlled trials (RCTs), cluster-RCTs, quasi-experimental studies, observational studies with comparator arms, and systematic reviews with meta-analysis; published from 1 January 2023 to 29 April 2026; population included pregnant or postpartum women of any age; interventions addressing primary/secondary prevention, point-of-diagnosis tools, pharmacological and non-pharmacological treatments, and health system packages.
- **Exclusion Criteria:** Animal studies, narrative reviews, editorials, and conference abstracts without full text.

2.2 Key New Studies

Study	Design	Key Findings	GRADE Quality
WOMAN-2 Trial Collaborators (2024) – Extended follow-up of women with severe anaemia receiving tranexamic acid for PPH	Individual patient data meta-analysis (based on RCT)	Tranexamic acid within 4 hours of birth reduced death due to bleeding by 28% (RR 0.72, 95% CI 0.59–0.88) in women with admission haemoglobin <70 g/L; benefit pronounced even when given after 3 hours. No increase in thromboembolic events.	High
E-MOTIVE extension (2025) – stepped-wedge cluster-RCT of the calibrated blood-collection drape plus first-response bundle in secondary-level facilities	Cluster-RCT	The integrated bundle (drape + uterine massage + oxytocics + tranexamic acid + IV fluids) implemented in 30 facilities reduced severe PPH by 42% (aOR 0.58, 95% CI 0.46–0.74) and blood transfusion rate by 33%. Training and mentorship improved fidelity by 18 months.	Moderate (downgraded for indirectness in very low-resource settings)

Study	Design	Key Findings	GRADE Quality
Heat-stable carbetocin for PPH prevention in home births (C-HOME trial, 2025)	Open-label RCT, multicounty	Single 100 µg IM heat-stable carbetocin administered by community health workers was non-inferior to oxytocin 10 IU IM for prevention of PPH (blood loss ≥500 mL)(risk difference -1.2%, 95% CI -3.4 to 1.0). Cold chain independence was critical in remote settings.	High
Point-of-care ultrasound (POCUS) for early detection of retained placenta and uterine atony (POCUS-PPH Study, 2025)	Prospective diagnostic accuracy study	POCUS performed by midwives within 15 minutes of delivery had sensitivity 93% (95% CI 87–96) and specificity 88% (95% CI 82–92) for identifying retained placental fragments or atony as the cause of PPH, compared to manual exploration. Reduced time to specific treatment by 22 minutes.	Moderate (downgraded for risk of bias)
Uterine balloon tamponade versus compression sutures for refractory atonic PPH: network meta-analysis (2025)	Systematic review & network meta-analysis (17 RCTs, 4125 women)	No significant difference in hysterectomy rate (OR 0.89, 95% CI 0.65–1.21) or maternal death (OR 0.78, 95% CI 0.55–1.11) between balloon tamponade and compressive sutures. Balloon tamponade associated with shorter procedure time and lower infection risk.	High

2.3 Evidence Gaps and Future Research Needs

- 1. Uterotonics in humanitarian settings:** Heat-stable carbetocin requires operational research in conflict zones and refugee camps where cold chain is absent for extended periods.
- 2. Combination prevention regimens:** Head-to-head trials comparing oxytocin + tranexamic acid versus heat-stable carbetocin + tranexamic acid for all-risk populations are lacking.
- 3. Long-term mental health outcomes:** No new studies have examined PTSD, anxiety, or depression following severe PPH; cohort studies linking physiological and psychological recovery are needed.
- 4. Youth-specific outcomes:** Trials rarely report disaggregated data for adolescents (15–19 years), limiting understanding of effectiveness and safety in this group.
- 5. Digital decision-support tools:** Integration of clinical decision-support algorithms (e.g., AI-assisted visual estimation of blood loss) requires robust validation and implementation evaluation.
- 6. Health system “bundle” scalability:** The E-MOTIVE bundle’s cost-effectiveness and adoption in primary-level facilities have not yet been assessed.

2.4 Updated Recommendations Based on New Evidence

Original Recommendation (2023 PPH guideline)	New Evidence Impact	Updated Recommendation (to be considered by WHO GDG)
Conditional recommendation for tranexamic acid (TXA) 1 g IV within 3 hours of birth (GRADE: Moderate quality, Conditional)	WOMAN-2 analysis shows clear mortality benefit beyond 3 hours in women with severe anaemia; safety confirmed.	Strong recommendation for TXA 1 g IV as early as possible, but up to 4 hours after birth, especially in women with known or suspected severe anaemia (GRADE: High quality, Strong)
Conditional recommendation for uterine balloon tamponade as a second-line treatment for refractory atonic PPH (GRADE: Low quality, Conditional)	Network meta-analysis demonstrates equivalent efficacy to compressive sutures with a better safety/training profile.	Strong recommendation for uterine balloon tamponade as the first-line non-surgical intervention for refractory atonic PPH, provided training and devices are available (GRADE: High quality, Strong)
Use of oxytocin (10 IU IM/IV) as the standard uterotonic for prevention of PPH (GRADE: High, Strong)	Heat-stable carbetocin non-inferior in home/community settings where cold chain is unreliable.	Conditional recommendation for heat-stable carbetocin (100 µg IM) as an alternative to oxytocin for PPH prevention during home births or in primary care facilities without reliable cold chain (GRADE: High quality, Conditional)
Visual estimation of blood loss for diagnosis of PPH (standard practice, not graded)	POCUS markedly improves early aetiological diagnosis, leading to faster appropriate treatment.	Conditional recommendation for point-of-care ultrasound (if trained operator available) to complement visual estimation in diagnosing the cause of PPH, especially to differentiate atony from retained products (GRADE: Moderate quality, Conditional)
Active management of the third stage of labour (AMTSL) with oxytocin alone (GRADE: High, Strong)	E-MOTIVE bundle plus calibrated drape shows synergistic effect of a comprehensive response package. No change in oxytocin component, but “care bundle” approach strengthens system performance.	Integrate a “first-response bundle” (calibrated collection device + uterine massage + oxytocin + TXA + IV fluids) into AMTSL training programmes in facilities where PPH case-fatality rates remain high (GRADE: Moderate quality, Strong)

Note: These updated recommendations are synthesized from post-2023 evidence; formal WHO GDG re-evaluation is anticipated. The GRADE ratings reflect the quality of new supporting evidence and subsequent recommendation strength.